CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

FLT3 / KIT: 11 molecules stopped early, all small molecules

14 stopped trials, but 11 molecules — all small molecules: Catequentinib, Pacritinib, Regorafenib, Dasatinib Anhydrous and 7 more. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

5.6%
14 of 252 closed trials (95% CI 3.3%–9.1%)

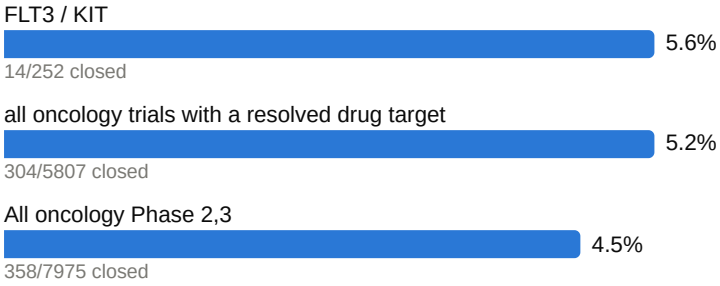
7 / 5 / 0
efficacy / safety / both (adds to 14)

33.4%
of 755 trials have closed · 503 still open or unresolved

1.6%
stopped for a biological reason within 3 years of starting (95% CI 0.7%–3.2%; 297 still at risk; 1.5% area-wide)

5 of 8 **Completed and missed the primary endpoint.** Of 172 completed trials in this segment, 8 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 5 missed and 3 met. Counted separately and never in the rate above; a floor, not a rate. 93 of the 162 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 311 distinct experimental drugs across 755 trials (small molecule 745, antibody 245, unknown 107, protein 30). Most tested: CATEQUENTINIB (260), REGORAFENIB (108), SORAFENIB (75), DASATINIB ANHYDROUS (47), SUNITINIB (35). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 14 stops came from 14 sponsor-asset programmes across 14 sponsors; the largest (Washington University School of Medicine) contributed 1; removing that programme's trials from both sides leaves 5.2%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 14 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.1%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	5	120	4.2%	1.8%–9.4%
2018–2020	6	80	7.5%	3.5%–15.4%
2021–2024	3	52	5.8%	2.0%–15.6%

Where the stops sit

Efficacy / futility: 7, Safety: 5, Benefit–risk: 2.
Sponsors with most stops: Great Novel Therapeutics Biotech & Medicals Corporation (1), Theseus Pharmaceuticals (1), Medical College of Wisconsin (1), Niguarda Hospital (1), First Affiliated Hospital of Zhejiang University (1), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05770882	2	Great Novel Therapeutics Biotech & Medicals Corporation	TUCIDINOSTAT, REGORAFENIB	2025-01	Benefit–risk	The overall profile does not support development for Hepatocellular Carcinoma

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05160168	2	Theseus Pharmaceuticals	Pan-variant KIT Inhibitor THE-630	2024-02	Safety	The Sponsor terminated the study due to early dose-limiting toxicities.
NCT04635059	2	Medical College of Wisconsin	PACRITINIB	2023-11	Efficacy	futility and drug supplier in favor of new metastatic protocol.
NCT03686345	2	Niguarda Hospital	MIDOSTAURIN	2023-06	Efficacy	didn't reach the primary endpoint
NCT04438902	2	First Affiliated Hospital of Zhejiang University	OSIMERTINIB, CATEQUENTINIB	2022-12	Safety	Many patients can't tolerate the combination treatment due to adverse events.
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.

The 6 most recent of 14 stopped trials. Every molecule behind all 14 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 77% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 33.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.